

Week 7: AC Power and Power Factor

Instantaneous and average power, complex power, power triangle, power factor correction, AC maximum power

In-depth notes covering all core material of this chapter: theory, engineering shortcuts, exam traps, 6 worked examples, and 8 practice problems with full solutions.

Based on: Rizzoni & Kearns, *Principles and Applications of Electrical Engineering*, 7th ed., §3.7–3.9

Inspired by: ELEC 275, Principles of Electrical Engineering, Concordia University

By **Vladislav Shkuratov**, made with Claude (AI). Independent study notes, not affiliated with Concordia University or any instructor.

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How to use this deck

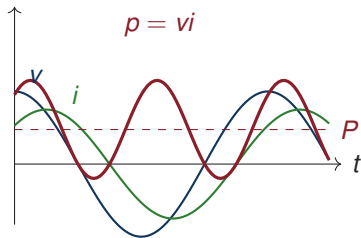
- **“Try it” slides:** the problem alone, with empty space. Do it first.
- **Next slide:** the clean solution, step by step.
- **Purple boxes:** engineering shortcuts, the fast way to see or solve things.
- **Orange boxes:** traps. This is where marks get lost on exams.
- Formulas are tagged **[on formula sheet]** or **[NOT on formula sheet]**, based on the sheet given in ELEC 275 exams.
- **End of deck:** final-check problems, then a full answer sheet.

The right side of each slide is left empty for your notes.

Symbols for this week

Symbol	Unit	Meaning
$p(t)$	W	instantaneous power = $v(t) i(t)$
P	W (watt)	average (real) power: what does useful work and shows on the bill
Q	var	reactive power: energy sloshing back and forth into L and C
$\mathbf{S} = P + jQ$	VA	complex power
$ \mathbf{S} $	VA	apparent power = $V_{rms} I_{rms}$: what the wires must carry
$\theta = \theta_v - \theta_i$	deg	power factor angle (the angle of the load's $\mathbf{Z}$)
$\text{pf} = \cos \theta$		power factor, between 0 and 1

Instantaneous power



With $v = V_m \cos(\omega t + \theta_v)$ and
 $i = I_m \cos(\omega t + \theta_i)$:

$$p(t) = \underbrace{\frac{1}{2} V_m I_m \cos \theta}_{\text{average } P} + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

$p(t)$ wiggles at **twice** the frequency around the average P . It can go **negative**: the load briefly sends energy back to the source.

Where the $p(t)$ formula comes from

Multiply the two waves and use the product-to-sum identity

$$\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]:$$

$$p = V_m \cos(\omega t + \theta_v) I_m \cos(\omega t + \theta_i) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

First term: a constant. It doesn't depend on time, so it is the **average** power P .

Second term: a cosine at 2ω . Over one cycle it averages to zero: energy that goes into the load and comes back out.

Only the phase **difference** $\theta = \theta_v - \theta_i$ matters for the average. That's why power problems only ever need the angle of the load's impedance.

Average power

[NOT on formula sheet: memorize]

$$P = \frac{1}{2} V_m I_m \cos \theta = V_{rms} I_{rms} \cos \theta, \quad \theta = \theta_v - \theta_i$$

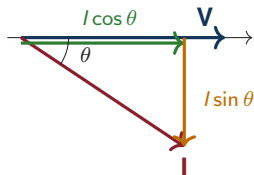
Peak or RMS: the factor of $\frac{1}{2}$

With **peak** values you need the $\frac{1}{2}$. With **RMS** values you don't, since $V_{rms} I_{rms} = \frac{1}{2} V_m I_m$. Power problems usually give RMS.

Engineering shortcut

Two faster routes: $P = I_{rms}^2 R$ (only the resistive part of **Z** burns power), or $P = \text{Re}\{\mathbf{S}\}$.

Why $\cos \theta$: only in-phase current works



Split the current into two parts:

$I \cos \theta$, in phase with V : behaves like current in a resistor. It delivers energy every cycle:
 $P = VI \cos \theta$.

$I \sin \theta$, 90° out of phase: behaves like current in an L or C. Energy goes in and comes back, averaging zero. It gives
 $Q = VI \sin \theta$.

Takeaway: the wires carry the full I , but only $I \cos \theta$ does useful work. That's why a low power factor is wasteful.

Try it: Example 1, average and reactive power

$$v(t) = 100 \cos(\omega t + 30^\circ) \text{ V}, \quad i(t) = 10 \cos(\omega t - 15^\circ) \text{ A (peak values)}$$

(a) θ (b) P (c) Q (d) the maximum of $p(t)$

your work here ↓

Example 1: solution

(a) θ power angle = voltage angle – current angle:

$$\theta = 30^\circ - (-15^\circ) = 45^\circ; \text{ positive, so the current lags (inductive)}$$

(b) P values are **peak**, so use the $\frac{1}{2}$:

$$P = \frac{1}{2} V_m I_m \cos \theta = \frac{1}{2} (100)(10) \cos 45^\circ = 353.6 \text{ W}$$

(c) Q same with sin: $Q = \frac{1}{2} (100)(10) \sin 45^\circ = 353.6 \text{ var}$
(positive: absorbed, inductive)

(d) $p(t) = P + \frac{1}{2} V_m I_m \cos(2\omega t + \dots)$; the cosine's biggest value is

$$1: p_{\max} = 353.6 + 500 = 853.6 \text{ W}$$

Shortcut: $p(t)$ swings between $P \pm |S|$ ($|S| = 500 \text{ VA}$ here).

Power in R, L and C

Element	θ	P	Q	Meaning
R	0°	$I_{rms}^2 R$	0	burns energy
L	$+90^\circ$	0	$+I_{rms}^2 X_L$	absorbs vars
C	-90°	0	$-I_{rms}^2 X_C $	supplies vars

Engineering shortcut

P comes only from resistors, and Q only from L and C. For any load:
 $P = I_{rms}^2 R$, $Q = I_{rms}^2 X$, using the R and X of its impedance.

Trap: an ideal L or C has **zero** average power, even with current flowing through it.

What reactive power really is

An inductor stores $\frac{1}{2}Li^2$ and a capacitor stores $\frac{1}{2}Cv^2$. With AC, that stored energy rises and falls twice every cycle: the element takes energy for a quarter cycle, then gives it all back the next quarter.

Q measures the size of that back-and-forth flow (in var, volt-ampere reactive). Its average is zero, so it does no useful work and costs nothing on a home bill, but the current that carries it still heats the wires.

Sign convention: inductors **absorb** vars ($Q > 0$), capacitors **supply** them ($Q < 0$). When both are present, they trade energy with each other instead of with the source. That's why a capacitor can cancel a motor's inductive vars.

Complex power and the power triangle

[NOT on formula sheet: memorize]

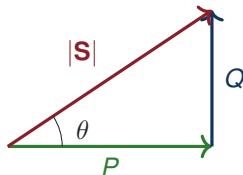
$$\mathbf{S} = \mathbf{V}_{rms} \mathbf{I}_{rms}^* = P + jQ$$

$$|\mathbf{S}| = V_{rms} I_{rms} \quad \text{apparent (VA)}$$

$$P = |\mathbf{S}| \cos \theta \quad \text{real (W)}$$

$$Q = |\mathbf{S}| \sin \theta \quad \text{reactive (var)}$$

$$|\mathbf{S}|^2 = P^2 + Q^2$$



power triangle (inductive load)

The conjugate

$\mathbf{S} = \mathbf{V}\mathbf{I}^*$, not $\mathbf{V}\mathbf{I}$. Forgetting the conjugate flips the sign of Q (inductive would look capacitive).

Why $S = VI^*$ needs the conjugate

Write $\mathbf{V} = V\angle\theta_v$ and $\mathbf{I} = I\angle\theta_i$ (RMS). We want a complex number whose angle is the **difference** $\theta_v - \theta_i$, because that's what sets P and Q .

$\mathbf{VI}$ $VI\angle(\theta_v + \theta_i)$ (angles **add**: meaningless)

$\mathbf{VI}^*$ $VI\angle(\theta_v - \theta_i) = VI\cos\theta + jVI\sin\theta = P + jQ$ ✓

The conjugate flips the current's angle, so multiplying *subtracts* it instead of adding it.

Handy forms for one load $\mathbf{Z}$

$\mathbf{S} = |\mathbf{I}|^2 \mathbf{Z} = \frac{|\mathbf{V}|^2}{\mathbf{Z}^*}$. So $\mathbf{S}$ always has the same angle as $\mathbf{Z}$: inductive $\mathbf{Z}$ gives $Q > 0$.

Try it: Example 2, complex power of a load

A load $\mathbf{Z} = 8 + j6 \, \Omega$ is connected to $120\angle 0^\circ$ V RMS.

Find $\mathbf{I}$, $\mathbf{S}$, P , Q , $|\mathbf{S}|$, and the power factor. Check P with $I^2 R$.

your work here ↓

Example 2: solution (1/2)

Z $8 + j6$: $\sqrt{64 + 36} = 10$, $\tan^{-1}(6/8) = 36.87^\circ$:
 $10\angle 36.87^\circ \Omega$

I Ohm: $\frac{120\angle 0^\circ}{10\angle 36.87^\circ} = 12\angle -36.87^\circ \text{ A}$ (RMS)

Conjugate $\mathbf{I}^* = 12\angle +36.87^\circ$ (flip the angle's sign)

S $\mathbf{VI}^* = (120)(12)\angle(0^\circ + 36.87^\circ) = 1440\angle 36.87^\circ =$
 $1152 + j864 \text{ VA}$

Example 2: solution (2/2)

Read off real part $P = 1152 \text{ W}$, imaginary $Q = 864 \text{ var}$, size

$$|S| = 1440 \text{ VA}$$

pf $\cos 36.87^\circ = 0.8$; $Q > 0$ (inductive), so 0.8 lagging

Check $I^2 R = 144 \times 8 = 1152 \text{ W} \checkmark$ $I^2 X = 144 \times 6 = 864 \text{ var} \checkmark$

Power factor: leading and lagging

[NOT on formula sheet: memorize]

$$\text{pf} = \cos \theta = \frac{P}{|S|}$$

Load	θ, Q	Current vs voltage	pf label
inductive (motors)	> 0	I lags V	lagging
capacitive	< 0	I leads V	leading
resistive	$= 0$	in phase	unity (1)

Why it matters: a low pf means more current for the same useful power, $I = P/(V \text{ pf})$: thicker wires, bigger losses, and utility penalties.

Conservation of complex power

Adding loads

The total complex power is the sum of each load's: $\mathbf{S}_{total} = \sum \mathbf{S}_k$, so P 's add and Q 's add (with signs). This works for series **and** parallel loads.

Trap

Apparent powers do **not** add: $|\mathbf{S}_{total}| \neq \sum |\mathbf{S}_k|$ in general. Add P and Q separately, then compute $|\mathbf{S}|$.

Engineering shortcut: loads given as “P at pf”

$Q = P \tan \theta$, with $\theta = \cos^{-1}(\text{pf})$. Lagging: $Q > 0$. Leading: $Q < 0$. No impedances needed.

Why complex power adds

Loads in **parallel** share $\mathbf{V}$, and the source current is $\mathbf{I} = \mathbf{I}_1 + \mathbf{I}_2$ (KCL). So

$$\mathbf{S} = \mathbf{V}\mathbf{I}^* = \mathbf{V}(\mathbf{I}_1 + \mathbf{I}_2)^* = \mathbf{V}\mathbf{I}_1^* + \mathbf{V}\mathbf{I}_2^* = \mathbf{S}_1 + \mathbf{S}_2$$

Loads in **series** share $\mathbf{I}$, and $\mathbf{V} = \mathbf{V}_1 + \mathbf{V}_2$ (KVL): the same algebra gives $\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2$.

Real and imaginary parts add separately, so the P 's add and the Q 's add. But $|\mathbf{S}_1 + \mathbf{S}_2| \neq |\mathbf{S}_1| + |\mathbf{S}_2|$ unless both loads have the same angle, just as two arrows pointing different ways don't add their lengths.

Try it: Example 3, two loads

Two loads in parallel on 240 V RMS:

Load 1: 10 kW at pf 0.8 lagging. Load 2: 6 kW at pf 0.6 leading.

Find the total **S**, the overall power factor, and the source current.

your work here ↓

Example 3: solution (1/2)

Plan loads given as “ P at pf”: find each Q from $Q = P \tan \theta$, add P 's and Q 's

Load 1 pf 0.8 lagging: $\theta = \cos^{-1} 0.8 = +36.87^\circ$ (lagging $\Rightarrow$ positive):
 $Q_1 = 10 \tan 36.87^\circ = +7.5 \text{ kvar}$

Load 2 pf 0.6 leading: $\theta = -\cos^{-1} 0.6 = -53.13^\circ$ (leading $\Rightarrow$ negative): $Q_2 = 6 \tan(-53.13^\circ) = -8 \text{ kvar}$

Total $P = 10 + 6 = 16 \text{ kW}$, $Q = 7.5 - 8 = -0.5 \text{ kvar}$:

$$\mathbf{S = 16 - j0.5 \text{ kVA}}$$

Example 3: solution (2/2)

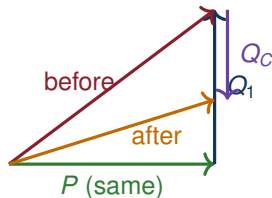
Size $|\mathbf{S}| = \sqrt{16^2 + 0.5^2} = 16.008 \text{ kVA}$

pf $P/|\mathbf{S}| = 16/16.008 = 0.9995$, and $Q < 0$: 0.9995 leading

Current $|\mathbf{S}| = VI$: $I = 16\,008/240 = 66.7 \text{ A}$

Power factor correction

[NOT on formula sheet: memorize]



Add a capacitor **in parallel** with the load. It supplies vars, shrinking Q . P doesn't change.

$$Q_C = P (\tan \theta_1 - \tan \theta_2)$$

$$C = \frac{Q_C}{\omega V_{rms}^2}$$

Trap

Use ω in rad/s and V_{rms} (not peak). The capacitor goes in **parallel**: in series it would change the load's voltage.

Where the correction formulas come from

Why a parallel capacitor: in parallel it sees the same voltage as the load, so the load works exactly as before. It supplies vars ($Q < 0$), which cancel part of the load's Q . P doesn't change: an ideal capacitor absorbs no real power.

How many vars: the power triangle gives $Q = P \tan \theta$. The capacitor supplies the difference between the old and the wanted Q :

$$Q_C = P \tan \theta_1 - P \tan \theta_2 = P(\tan \theta_1 - \tan \theta_2), \quad \theta = \cos^{-1}(\text{pf})$$

Which capacitor: with V_{rms} across it, a capacitor supplies

$|Q_C| = V_{rms}^2 / |X_C| = V_{rms}^2 \omega C$. Solve for C :

$$C = Q_C / (\omega V_{rms}^2)$$

Try it: Example 4, power factor correction

A plant draws 50 kW at pf 0.8 lagging from a 480 V RMS, 60 Hz line.

- (a) What parallel capacitor raises the pf to 0.95 lagging?
- (b) Line current before and after?

your work here ↓

Example 4: solution (1/2)

Before $\theta_1 = \cos^{-1} 0.8 = 36.87^\circ$: $Q_1 = 50 \tan 36.87^\circ = 37.5 \text{ kvar}$

Target $\theta_2 = \cos^{-1} 0.95 = 18.19^\circ$: $Q_2 = 50 \tan 18.19^\circ = 16.43 \text{ kvar}$

Capacitor it must supply the difference: $Q_C = 37.5 - 16.43 = 21.07 \text{ kvar}$

Example 4: solution (2/2)

(a) C
$$C = \frac{Q_C}{\omega V_{rms}^2} = \frac{21\,070}{(2\pi \cdot 60)(480^2)} = \frac{21\,070}{(377)(230\,400)} =$$

 $242.5 \mu\text{F}$

(b) before
$$I = \frac{P}{V_{pf}} = \frac{50\,000}{480 \times 0.8} = 130.2 \text{ A}$$

(b) after
$$I = \frac{50\,000}{480 \times 0.95} = 109.6 \text{ A} \quad (\text{same } P, 16\% \text{ less current})$$

AC maximum power transfer

[NOT on formula sheet: memorize]

Load can be any impedance: conjugate match

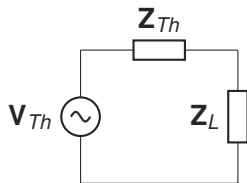
$$\mathbf{Z}_L = \mathbf{Z}_{Th}^* \text{ (same } R, \text{ opposite } X). \text{ Then } P_{max} = \frac{|\mathbf{V}_{Th,rms}|^2}{4R_{Th}}.$$

Load must be a pure resistor

$$R_L = |\mathbf{Z}_{Th}| = \sqrt{R_{Th}^2 + X_{Th}^2}. \text{ The power is lower than the conjugate match.}$$

Why: the conjugate cancels the reactance ($X_{Th} + X_L = 0$), leaving the week 3 problem with $R_L = R_{Th}$.

Deriving AC maximum power (1/2)



$$\begin{aligned} \mathbf{Z}_{Th} &= R_{Th} + jX_{Th} \\ \mathbf{Z}_L &= R_L + jX_L \end{aligned}$$

One loop, so

$$|\mathbf{I}| = \frac{|\mathbf{V}_{Th}|}{\sqrt{(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2}}$$

Only the load's resistance absorbs power:

$$P_L = |\mathbf{I}|^2 R_L = \frac{|\mathbf{V}_{Th}|^2 R_L}{(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2}$$

Step 1, the reactance: X_L only appears in the bottom, inside a square. The bottom is smallest (and P_L biggest) when $X_{Th} + X_L = 0$, i.e. $X_L = -X_{Th}$.

Step 2, the resistance: with the reactances cancelled, $P_L = \frac{|\mathbf{V}_{Th}|^2 R_L}{(R_{Th} + R_L)^2}$:

exactly the week 3 problem, so $R_L = R_{Th}$.

Deriving AC maximum power (2/2)

Result (conjugate match): $\mathbf{Z}_L = R_{Th} - jX_{Th} = \mathbf{Z}_{Th}^*$, and

$$P_{max} = \frac{|\mathbf{V}_{Th,rms}|^2 R_{Th}}{(2R_{Th})^2} = \frac{|\mathbf{V}_{Th,rms}|^2}{4R_{Th}}$$

If the load must be a pure resistor ($X_L = 0$, can't cancel anything): the bottom becomes $(R_{Th} + R_L)^2 + X_{Th}^2$. Setting $dP_L/dR_L = 0$ gives

$$R_L = \sqrt{R_{Th}^2 + X_{Th}^2} = |\mathbf{Z}_{Th}|$$

This gives less power than the conjugate match, because the uncanceled X_{Th} limits the current. **Trap:** use the **RMS** Thévenin voltage in P_{max} . With a peak value, divide by $8R_{Th}$ instead of $4R_{Th}$.

Try it: Example 5, AC maximum power

A network has $\mathbf{V}_{Th} = 10\angle 0^\circ$ V RMS and $\mathbf{Z}_{Th} = 4 + j3 \Omega$.

- (a) Load impedance for maximum power, and that power.
- (b) If the load must be a pure resistor: which R_L , and what power?

your work here ↓

Example 5: solution (1/2)

(a) **load** conjugate match: same R , opposite X : $\mathbf{Z}_L = 4 - j3 \Omega$

(a) **power** reactances cancel, loop is $4 + 4 = 8 \Omega$ resistive:

$$P_{max} = \frac{|\mathbf{V}_{Th}|^2}{4R_{Th}} = \frac{10^2}{16} = 6.25 \text{ W}$$

(b) **load** resistor only: $R_L = |\mathbf{Z}_{Th}| = \sqrt{4^2 + 3^2} = 5 \Omega$

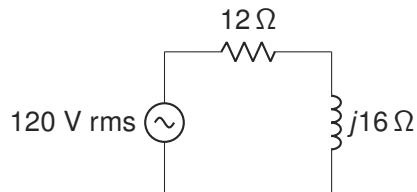
Example 5: solution (2/2)

(b) current loop impedance $(4 + 5) + j3 = 9 + j3$, size
 $\sqrt{81 + 9} = 9.49$: $|I| = 10/9.49 = 1.054 \text{ A}$

(b) power only R_L absorbs: $P = |I|^2 R_L = 1.054^2 \times 5 = 5.56 \text{ W}$

Compare $5.56 \text{ W} < 6.25 \text{ W}$: the uncanceled $j3$ limits the current

Try it: Example 6, exam style



60 Hz supply. Find:

- (a) I , P , Q , $|\mathbf{S}|$, pf
- (b) the parallel C for unity pf
- (c) the line current after correction

your work here ↓

Example 6: solution (1/2)

(a) I $|Z| = |12 + j16| = \sqrt{144 + 256} = 20 \Omega$:

$$I = 120/20 = 6 \text{ A rms}$$

(a) P, Q $P = I^2 R = 36 \times 12 = 432 \text{ W}$, $Q = I^2 X = 36 \times 16 = 576 \text{ var}$

(a) $|S|, \text{ pf}$ $|S| = VI = 720 \text{ VA}$ ($= \sqrt{432^2 + 576^2} \checkmark$), $\text{pf} = 432/720 = 0.6 \text{ lagging}$

Example 6: solution (2/2)

(b) unity pf means $Q_{total} = 0$, so the capacitor supplies all 576 var:

$$C = \frac{576}{(377)(120^2)} = 106 \mu\text{F}$$

(c) now $|\mathbf{S}| = P = 432 \text{ VA}$: $I = 432/120 = 3.6 \text{ A}$ (was 6 A)

Energy cost and mixed frequencies

Energy and cost

The bill is for real energy: $\text{kWh} = P \text{ (kW)} \times \text{hours}$. $\text{Cost} = \text{kWh} \times \text{rate}$.

Mixed frequencies (and DC)

When sources run at **different** frequencies, the powers **do** add: $P = P_{DC} + P_{\omega_1} + P_{\omega_2} + \dots$

Equivalently, $I_{rms} = \sqrt{I_{DC}^2 + I_{1,rms}^2 + \dots}$

Trap: powers from sources at the **same** frequency do **not** add. Add the phasors first.

Why powers at different frequencies add

Take $i = I_1 \cos \omega_1 t + I_2 \cos \omega_2 t$ through a resistor. Then

$$i^2 R = (I_1^2 \cos^2 \omega_1 t + I_2^2 \cos^2 \omega_2 t + 2I_1 I_2 \cos \omega_1 t \cos \omega_2 t) R$$

The cross term $2I_1 I_2 \cos \omega_1 t \cos \omega_2 t$ splits into cosines at $\omega_1 \pm \omega_2$, and each averages to **zero** when $\omega_1 \neq \omega_2$. Only the two squared terms survive:

$$P = P_1 + P_2 \quad I_{rms}^2 = I_{1,rms}^2 + I_{2,rms}^2$$

At the **same** frequency the cross term does **not** average to zero (it depends on the phase difference). That's why same-frequency powers can't simply be added: add the phasors first.

Traps

Check these before you box an answer

- Peak values need $\frac{1}{2}$ in P . RMS values don't.
- $\mathbf{S} = \mathbf{VI}^*$: conjugate the **current**.
- $\theta = \theta_v - \theta_i$, the angle of $\mathbf{Z}$, not of the current alone.
- Lagging pf $\rightarrow Q > 0$ (inductive). Leading $\rightarrow Q < 0$.
- Correction capacitor: parallel, V_{rms} , ω in rad/s, P unchanged.
- Add P and Q separately. Never add $|\mathbf{S}|$ values.

Exam-format traps

Units

P in W, Q in var, $|\mathbf{S}|$ in VA. Using W for all three costs marks.

Say lagging or leading

A power factor of “0.8” alone is incomplete. Write “0.8 lagging”.

Draw the power triangle

For pf correction, a quick sketch of the before and after triangles shows your method and catches sign errors.

Summary (1/2)

Average power $P = V_{rms} I_{rms} \cos \theta = \frac{1}{2} V_m I_m \cos \theta = I_{rms}^2 R$

Reactive power $Q = V_{rms} I_{rms} \sin \theta = I_{rms}^2 X$

Complex power $\mathbf{S} = \mathbf{V}_{rms} \mathbf{I}_{rms}^* = P + jQ, |\mathbf{S}| = V_{rms} I_{rms}$

Elements R: P only; L: $+Q$; C: $-Q$

Summary (2/2)

Power factor	$\text{pf} = \cos \theta = P/ \mathbf{S} $; lagging (inductive) or leading (capacitive)
Adding loads	P 's add, Q 's add; $Q = P \tan(\cos^{-1} \text{pf})$
Correction	$Q_C = P(\tan \theta_1 - \tan \theta_2)$, $C = Q_C/(\omega V_{rms}^2)$
Max power	$\mathbf{Z}_L = \mathbf{Z}_{Th}^*$, $P_{max} = V_{Th,rms}^2/4R_{Th}$; resistive only: $R_L = Z_{Th} $

Final check

8 problems. One per slide, with space. No looking back.

If one takes more than 10 min, go back to that section. The answer sheet comes after the last problem.

P1: average and reactive power

$$v(t) = 170 \cos(377t) \text{ V}, \quad i(t) = 5 \cos(377t - 60^\circ) \text{ A (peak)}.$$

Find P , Q , $|\mathbf{S}|$ and the power factor.

your work here ↓

P2: capacitive load

$\mathbf{Z} = 30 - j40 \, \Omega$ on $200\angle 0^\circ \text{ V RMS}$.

Find I , P , Q , and the power factor (with its label).

your work here ↓

P3: three loads

On 480 V RMS: a motor (20 kW, pf 0.8 lagging), a heater (10 kW, pf 1) and a capacitor bank (5 kvar).

Find the total P , Q , the overall pf and the line current.

your work here ↓

P4: power factor correction

A 100 kW load at pf 0.7 lagging is fed from 600 V RMS, 60 Hz.

Find the parallel capacitance that raises the pf to 0.9 lagging.

your work here ↓

P5: AC maximum power

$\mathbf{V}_{Th} = 20\angle 0^\circ$ V RMS, $\mathbf{Z}_{Th} = 6 - j8 \Omega$.

(a) Best load and P_{max} . (b) Best pure resistor and its power.

your work here ↓

P6: instantaneous power

For P1, find the maximum and minimum of $p(t)$. What does the negative value mean?

your work here ↓

P7: energy cost

A 1.5 kW heater runs 4 hours a day for 30 days. Electricity costs \$0.08 per kWh.

How much energy is used, and what does it cost?

your work here ↓

P8: exam style

A load $\mathbf{Z} = 16 + j12 \Omega$ is fed from 240 V RMS, 60 Hz.

(a) I , P , Q , pf. (b) The parallel capacitor for pf 0.95 lagging.

your work here ↓

Answer sheet

Full worked solutions to P1–P8.

P1: solution

θ $\theta_v - \theta_i = 0^\circ - (-60^\circ) = 60^\circ$ (current lags: inductive)

P peak values, so the $\frac{1}{2}$: $\frac{1}{2}(170)(5) \cos 60^\circ = 425 \times 0.5 =$
212.5 W

Q $\frac{1}{2}(170)(5) \sin 60^\circ = 425 \times 0.866 =$ **368.1 var**

$|S|$ $\frac{1}{2}(170)(5) =$ **425 VA** (check: $\sqrt{212.5^2 + 368.1^2} = 425 \checkmark$)

pf $\cos 60^\circ = 0.5$, current lags: **0.5 lagging**

P2: solution

I $|Z| = \sqrt{30^2 + 40^2} = 50$: $I = 200/50 = 4 \text{ A}$ (RMS)

P only the resistance: $I^2 R = 16 \times 30 = 480 \text{ W}$

Q $I^2 X$ with $X = -40$: $16 \times (-40) = -640 \text{ var}$ (negative: capacitive, it supplies vars)

pf $|S| = VI = 800 \text{ VA}$, $\text{pf} = 480/800 = 0.6$; current leads (capacitive): 0.6 leading

P3: solution (1/2)

Plan add every load's P and Q separately, then find pf and current from the totals

Motor pf 0.8 lagging: $Q = 20 \tan(\cos^{-1} 0.8) = 20 \times 0.75 = +15$ kvar

Heater pf 1: pure resistance, $Q = 0$

Capacitors supply vars: $Q = -5$ kvar, $P = 0$

P3: solution (2/2)

Totals $P = 20 + 10 + 0 = 30 \text{ kW}$, $Q = 15 + 0 - 5 = 10 \text{ kvar}$

pf $|\mathbf{S}| = \sqrt{30^2 + 10^2} = 31.62 \text{ kVA}$: $30/31.62 =$
 0.949 lagging ($Q > 0$)

Current $I = |\mathbf{S}|/V = 31\,623/480 = 65.9 \text{ A}$

P4: solution

Before $\theta_1 = \cos^{-1} 0.7 = 45.57^\circ$: $Q_1 = 100 \tan 45.57^\circ = 102.0$
kvar

Target $\theta_2 = \cos^{-1} 0.9 = 25.84^\circ$: $Q_2 = 100 \tan 25.84^\circ = 48.4$ kvar

Capacitor $Q_C = 102.0 - 48.4 = 53.6$ kvar

$$C \quad \frac{Q_C}{\omega V_{rms}^2} = \frac{53\,590}{(377)(600^2)} = \boxed{395 \mu\text{F}}$$

P5: solution

(a) **load** conjugate: same R , opposite X : $\mathbf{Z}_L = 6 + j8 \Omega$

(a) **power** $\frac{|\mathbf{V}_{Th}|^2}{4R_{Th}} = \frac{20^2}{4 \times 6} = 16.7 \text{ W}$

(b) **load** resistor only: $R_L = |\mathbf{Z}_{Th}| = \sqrt{36 + 64} = 10 \Omega$

(b) **current** loop: $(6 + 10) - j8 = 16 - j8$, size 17.89:
 $|\mathbf{I}| = 20/17.89 = 1.118 \text{ A}$

(b) **power** $1.118^2 \times 10 = 12.5 \text{ W}$ (less than 16.7 W: the $-j8$ isn't cancelled)

P6: solution

Formula $p(t) = P + |\mathbf{S}| \cos(2\omega t + \dots)$, and a cosine swings between +1 and -1

Max $P + |\mathbf{S}| = 212.5 + 425 = 637.5 \text{ W}$

Min $P - |\mathbf{S}| = 212.5 - 425 = -212.5 \text{ W}$

Meaning negative p : for part of each cycle the load's inductance gives stored energy **back** to the source

P7: solution

Energy energy = power $\times$ time, in kW and hours:

$$1.5 \times (4 \times 30) = 1.5 \times 120 = 180 \text{ kWh}$$

Cost $180 \times \$0.08 = \14.40

Note the bill counts real energy only (P). Reactive power doesn't appear on a home bill

P8: solution

(a) I $|\mathbf{Z}| = \sqrt{16^2 + 12^2} = 20$: $I = 240/20 = 12 \text{ A}$

(a) P, Q $I^2 R = 144 \times 16 = 2304 \text{ W}$, $I^2 X = 144 \times 12 = 1728 \text{ var}$

(a) pf $R/|\mathbf{Z}| = 16/20 = 0.8 \text{ lagging}$

(b) **target** $Q_2 = 2304 \tan(\cos^{-1} 0.95) = 2304 \times 0.3287 = 757.3 \text{ var}$

(b) **cap** $Q_C = 1728 - 757.3 = 970.7 \text{ var}$, $C = \frac{970.7}{(377)(240^2)} = 44.7 \mu\text{F}$